

factors, and complexes in and out of the nucleus to maintain its function and integrity. Small molecules, ions, and proteins up to 45,000 daltons passively diffuse through the pores. However, the vast majority of material transported is through a highly controlled process called “gating,” which is responsible for keeping complexes such as the ribosomes in the cytoplasm from entering the nucleus.

Some proteins require multiple crossings through the nuclear pore. Ribosomal proteins are first made in the cytoplasm, transported into the nucleus, assembled into ribosome subunits by the nucleolus, and then transported back out into the cytoplasm. Viruses infect nuclei by taking advantage of the presence of nuclear pores. Some can be transported intact, while others “dock” on the cytoplasmic side of the pore and inject their DNA into the nucleus through the pore’s opening. Each nuclear pore can both import and export material in one of two ways.

Any protein transported in or out of the nucleus must contain a nuclear localization signal, which is a specific sequence of four to eight amino acids that triggers either nuclear import or export. Each nuclear pore contains nucleoporins that recognize either the import or export signal, called importins or exportins, respectively. Importins, located on the cytoplasmic side of the nuclear pore, bind their import “cargo” and flip or slide it to the inside of the nucleus. They then move back into their original position, ready to “transport” their next “cargo.” The opposite happens on the side of the nuclear pore facing the interior of the nucleus. Here, exportins bind proteins within the nucleus carrying the export signal and flip or slide them through the pore and into the cytoplasm. RNA molecules and complexes can also move through the pores, but only if the importins and or exportins recognize them as cargo. **SEE ALSO** CELL CYCLE; CHROMOSOME, EUKARYOTIC; MUSCULAR DYSTROPHY; PROTEIN TARGETING; RIBOSOME; TELOMERE.

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Oncogenes

An oncogene is a gene that causes cancer. Oncogenes arise from normal cellular genes, often ones that help regulate cell division.

Early Oncogene Research

The first clues that cancer has a genetic basis came from several independent observations. In 1914 the German cell biologist Theodor Boveri viewed cancer cells through a microscope and noted that they often carried abnormal chromosomes. However, recognition that a specific chromosomal abnormality was routinely associated with a particular type of cancer did not come until 1973, when Janet Rowley showed that chronic myelogenous leukemia (CML) cells carried a chromosomal translocation in which the ends of chromosomes nine and twenty-two are exchanged. Several other studies showed that certain types of cancer can run in families, suggesting that cancer risk can be inherited. Then, in 1981 the laboratories of Robert Weinberg, Michael Wigler, Geoff Cooper, and Mariano Barbacid showed that DNA from a human bladder cancer cell line could cause nonmalignant cells in tissue culture to become cancerous.

Since the Weinberg and Wigler observations, dozens of oncogenes have been identified and characterized. It is clear that oncogenes represent certain normal cellular genes that are aberrantly expressed or functionally abnormal. Such normal cellular genes, or “proto-oncogenes,” can be altered to become oncogenes through a variety of different molecular mechanisms.

RNA Tumor Viruses and Proto-Oncogenes

In 1911 Peyton Rous reported that a class of RNA viruses can cause tumors in animals. These RNA tumor viruses, called “retroviruses,” carry an RNA genome that, once inside a cell, is copied into DNA, which then is inserted randomly into the **genome** of a host cell. Some retroviruses are slow to cause tumors. After infection and spread to a large number of cells, a DNA copy of the viral genome, by chance, integrates into a host cell’s DNA next to a normal gene that plays an important role in cell growth. If this viral integration disrupts the expression or structure of the normal cellular gene, it induces abnormal growth signals that can lead to cancer.

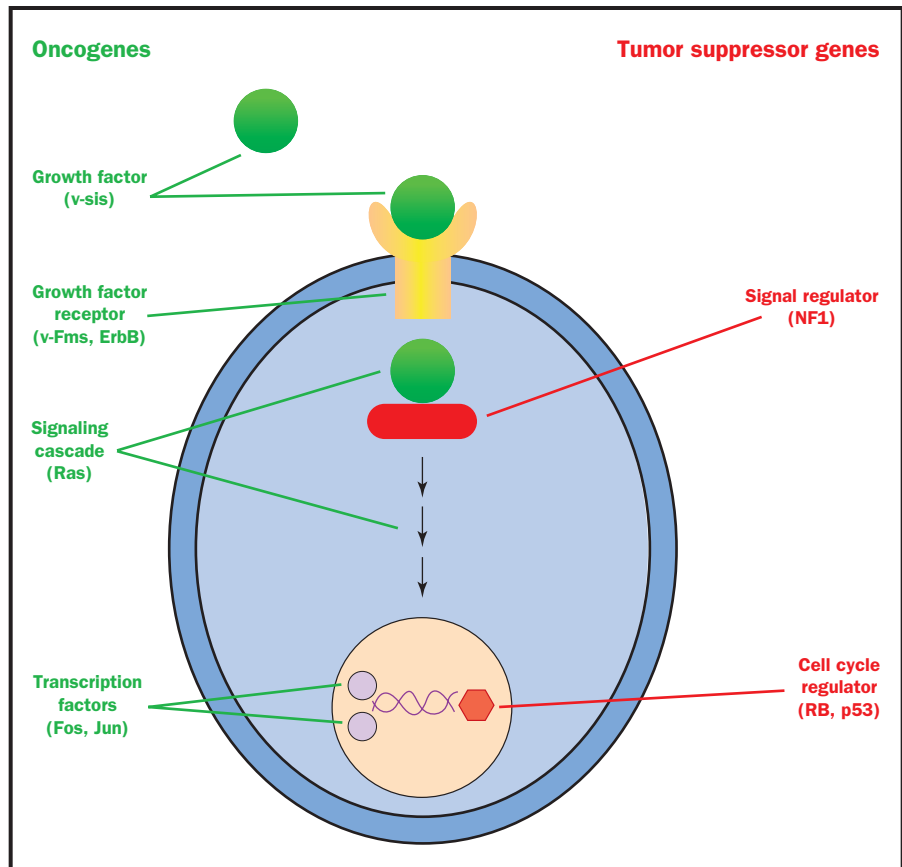
Other retroviruses cause tumors to appear very quickly. In the process of copying viral RNA into DNA, RNA that is expressed from cellular genes can be mistakenly copied into the viral genome. The **progeny** of the virus transfer the cellular gene to many other cells. If this “captured” cellular RNA is from a gene that stimulates cell growth, it then causes abnormal growth stimulation, leading to cancer. This process is termed “gene capture.”

Through molecular cloning, the genes that are activated or captured by retroviruses have been identified and characterized. Almost three dozen such retroviral oncogenes and their related cellular proto-oncogenes are now known.



genome the total genetic material in a cell or organism

progeny offspring

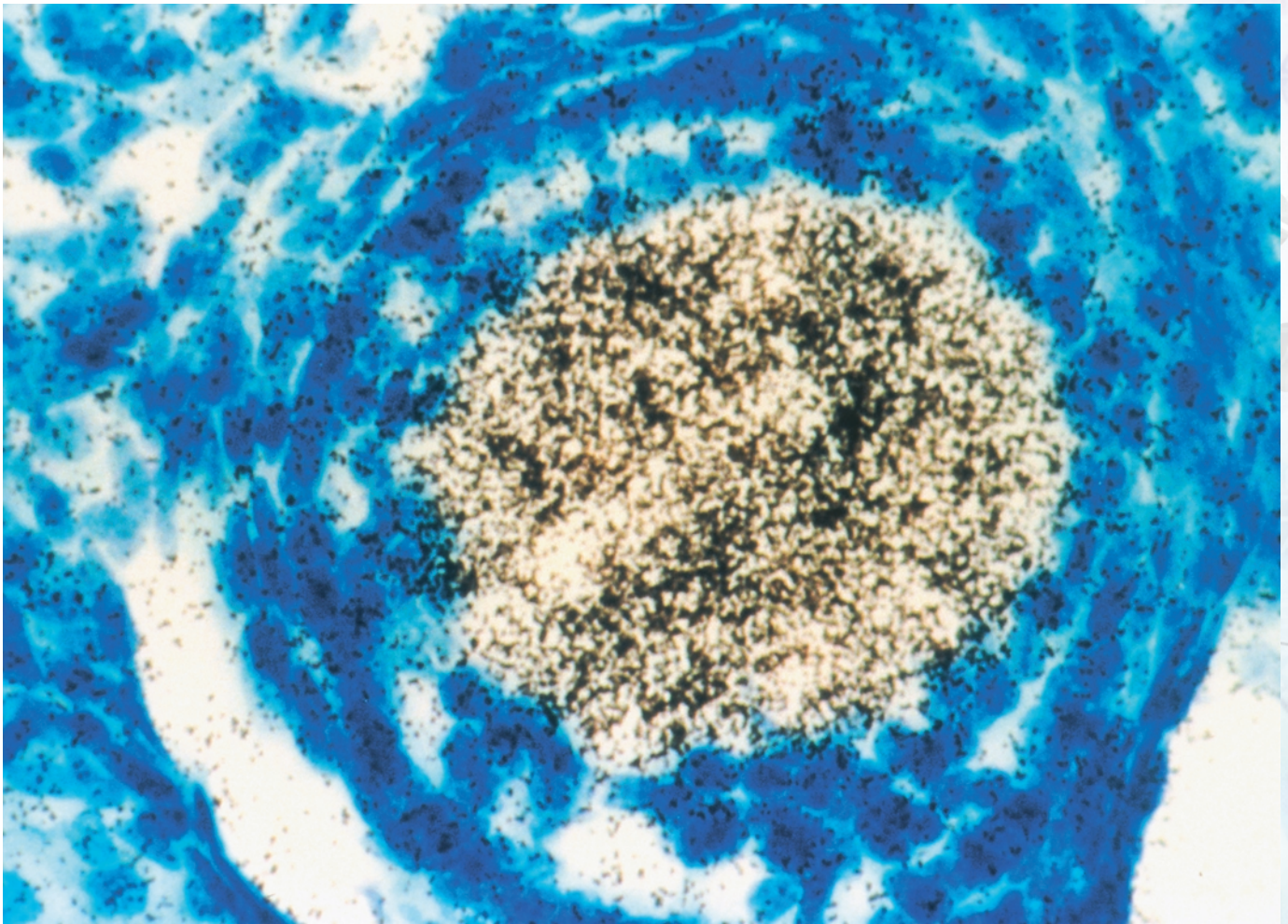


Oncogenes and tumor suppressor genes collaborate to “short circuit” normal growth regulatory pathways in cells. A generic growth-signaling pathway is shown where a growth factor binds to its receptor, which stimulates a cascade of signals inside the cell. The signals culminate in the nucleus, where they induce expression of genes involved in stimulating cell proliferation. On the left side are different steps in the signaling pathway that can be affected by activation of oncogenes. On the right side are growth inhibitory steps that can be lost by inactivation of tumor suppressor genes. In parentheses are examples of oncogenes and tumor suppressor genes that are mentioned in the text.

Proto-Oncogene Activation without Retroviruses

The first human oncogene, called *Ras*, was identified in the Weinberg and Wigler experiments. The protein product of the *Ras* gene serves as a switch that turns growth signals on and off. Normally, the activity of this *Ras* switch is tightly regulated. However, single mutations (called point mutations) in critical sites of the *Ras* gene cause the *Ras* growth switch to remain constantly turned on, which contributes to cancer. Thus, some proto-oncogenes can become oncogenes by genetic point mutation. Such mutations are not dependent upon the presence of retroviruses. Instead, they can occur during normal cell division and can be caused by environmental factors such as chemicals, ultraviolet rays from the sun, and X rays. DNA repair mechanisms usually, but not always, correct such mutations.

Oncogenes also can be activated by structural changes, called “amplifications” or “translocations,” that occur in the chromosomes. DNA amplification increases by several-fold a specific region of a chromosome. This can produce many copies of any genes that lie in the amplified region. If one of the genes in this region is important in driving cell growth, its over-expression due to amplification leads to uncontrolled proliferation.



In the case of chromosome translocations, a proto-oncogene on one chromosome might be moved to another chromosome, resulting in the gene's structural alteration and/or aberrant expression. For example, in the translocation between chromosomes 9 and 22 that is found in CML, a proto-oncogene on chromosome 9, called *c-Abl*, is moved to chromosome 22, where it is fused to another gene called *Bcr*. Normally, *c-Abl* is a nuclear enzyme called "tyrosine kinase," which adds a phosphate molecule to proteins at an amino acid called tyrosine. **Phosphorylation** regulates the function of certain proteins that play important roles in stimulating cell proliferation. The fusion of *Bcr* and *c-Abl* genes creates an oncogene, called *Bcr/Abl*, which makes a highly overactive tyrosine kinase variant that is found in the cytoplasm instead of the nucleus. These changes in the activity and cellular location of the *c-Abl* proto-oncogene lead to chronic myelogenous leukemia.

In situ hybridization reveals that the product of a known oncogene, inserted into a mouse egg cell, hybridizes to multiple copies of RNA (black spots).

phosphorylation addition of the phosphate group PO_4^{3-}

Short-Circuiting Normal Cell Growth Mechanisms

Normal cell growth is controlled by the availability of growth factors, which are hormone-like molecules that bind to specific receptors embedded in the surface membrane of cells. When this happens, the receptor stimulates a signaling cascade inside cells that ultimately tells the cells to divide. Many gene products in this signaling pathway are proto-oncogenes that can become oncogenes when activated by the different mechanisms described

above. When a signaling proto-oncogene is activated, the signaling cascade becomes “short-circuited” and cells behave as if they are continually stimulated by their growth factor.

For example, the *v-sis* oncogene from a monkey cancer virus known as simian sarcoma retrovirus (SSV) comes from a gene that encodes platelet-derived growth factor, which stimulates growth of different cell types. Cells infected with SSV are, therefore, constantly bathed in the *v-sis* growth factor and stimulated to proliferate. Other oncogenes are mutated growth factor receptors where mutation leaves the receptor in the “on” status even in the absence of the growth factor. Two examples of mutated receptor oncogenes include *v-erbB*, found in a bird retrovirus that causes various cancers, and *v-fms*, which is carried by a mouse retrovirus that causes leukemia.

Inside the cell, components of the signaling cascade that connect cell surface growth receptors to the nucleus also can cause cancer when their activity is altered by mutation or overexpression. The *Ras* proto-oncogene is an example of a signal-transmitting molecule inside cells that can mutate into an oncogene. In the nucleus, these normal growth signals trigger other proteins, called transcription factors, that regulate gene expression needed for cell growth. Many transcription factors are proto-oncogenes. Two examples of proto-oncogene transcription factors are *c-Fos* and *c-Jun*, both of which were first identified as retroviral oncogenes.

Tumor Suppressor Genes

In 1983 Raymond White and Webster Cavanee, using a technique called chromosome mapping, learned that a loss of a small segment of human chromosome 13 was a recurring feature in retinoblastoma, a rare childhood cancer of the retina that can run in families. In this deleted region they discovered a gene called *RB* (for retinoblastoma), both copies of which are inactivated either by DNA deletion or by a mutation within the gene that destroys its function. Such inactivation of both copies of the *RB* gene occurs in about 40 percent of human cancers. The product of the normal *RB* gene functions as a brake to cell division, so that loss of this brake can lead to unregulated cell growth.

Another gene associated with cancer when both copies are affected by mutation is the *p53* gene, which acts as a “guardian” of the genome. Normally, this gene product induces a cell suicide program called apoptosis in cells with damaged DNA. Loss of *p53* activity allows cells with damaged DNA to grow and pass DNA mutations to their daughter cells. A third type of gene that plays a role in cancer when it is inactivated is *NF1*. This gene encodes a protein that turns off the *Ras* growth signal mentioned above. In this case, loss of *NF1* function is another way that the *Ras* signal can be left constantly on.

Since the discovery of *RB*, researchers have identified several additional genes in which both copies are inactivated due to mutation or chromosomal deletion. These genes normally block cell growth; hence they are called tumor suppressor genes. Since both copies of these genes need to be inactivated in order to release cancer cells from growth inhibition, tumor suppressor genes act recessively. This contrasts to the oncogenes described above, where only one copy needs to be activated in order to promote cancer. Oncogenes, therefore, act in a dominant fashion.

Multiple Genetic “Hits”

In 1971 Alfred Knudson Jr. proposed that retinoblastoma resulted from at least two separate genetic defects. In families with a high risk of retinoblastoma, the first defect is inherited and the second occurs sometime during childhood. This came to be known as Knudson’s “two-hit theory.” Subsequent research has shown that most, if not all, cancer arises from multiple genetic events, or “hits.”

In many cancers, more than two hits are required. Bert Vogelstein and coworkers first showed this in colon cancer in the late 1980s. Colon cancer begins with a precancerous stage, called a benign polyp. Left untreated, this will progress through successively more cancerous stages until it becomes an aggressive carcinoma. Vogelstein’s group found that progression of colon cancer through these different stages was associated with the acquisition of genetic changes in oncogenes such as *Ras*, as well as in a number of different tumor suppressor genes, including *p53*. Together, sequential activation of different oncogenes along with inactivation of various tumor suppressor genes drive the step-wise progression of precancerous cells to highly malignant tumors. SEE ALSO APOPTOSIS; BREAST CANCER; CANCER; CARCINOGENS; CELL CYCLE; COLON CANCER; INHERITANCE PATTERNS; MUTATION; RETROVIRUS; SIGNAL TRANSDUCTION; TUMOR SUPPRESSOR GENES; TRANSCRIPTION FACTORS.

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Operon

An operon is a genetic regulatory system found in prokaryotes and the bacterial viruses (**bacteriophages**) that attack bacteria. It is a cluster of genes that share regulatory elements and are usually functionally related.

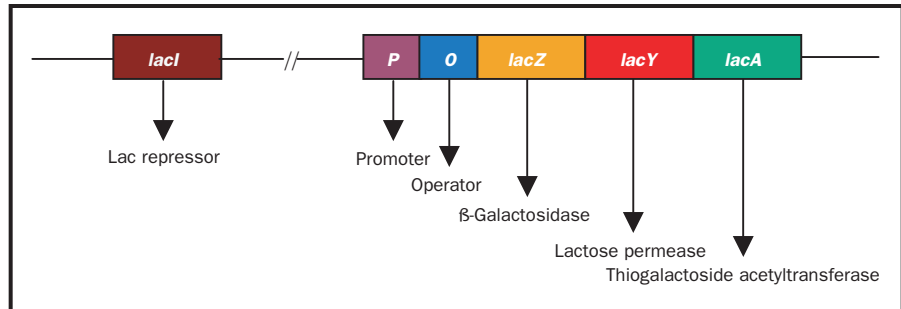
bacteriophages viruses that infect bacteria

The Discovery of Operons

French scientists Jacques Monod and François Jacob first coined the term “operon” in a short paper published in 1960 in the *Proceedings of the French Academy of Sciences*. They elaborated the concept of the operon in several papers that appeared in 1961, based on their studies on the *lac* genes (genes for the **metabolism** of lactose sugar) of the bacterium *Escherichia coli* and the genes of bacteriophage lambda. Monod and Jacob received the Nobel Prize in 1985 for this work.

metabolism chemical reactions within a cell

Figure 1. The *lac* operon. A single straight line signifies the double-stranded DNA molecule. The *lac* operon includes genes and control elements which are needed for metabolism of lactose.



Typical Features of Operons

The genes of an operon are usually functionally related. Genes are the basic unit of biological information, and consist of specific segments of deoxyribonucleic acid (DNA). The segments of DNA that constitute a gene consist of distinctive sets of nucleotide pairs located in a discrete region of a chromosome that encodes a particular protein. Within an operon, the genes encode proteins that execute related functions. For example, the five genes of the tryptophan (*trp*) operon in *E. coli* each encode one of the enzymes necessary for the biosynthesis of the amino acid tryptophan from a metabolite called chorismate. This condition is mimicked in many bacteria.

Exceptions do occur; the genes of some operons may lack an obvious functional relationship. For example one operon in *E. coli* contains one gene that encodes a ribosomal protein S21 (*rpsU*), another that encodes DNA primase (*dnaG*), and one that encodes the sigma subunit of RNA polymerase (*rpoD*). The protein products of these genes are all involved in starting up the **synthesis of macromolecules**, but beyond that they have no obvious functional relationships to one another. Nonetheless, the clustering of these genes and their common regulation qualify them to be treated as elements of a single operon.

Another common feature of operons is that their genes are clustered on the bacterial chromosome. This chromosome is a large circular molecule of DNA. The genes of an operon are arranged in a consecutive and linear fashion at a specific location on the bacterial chromosome.

In the case of the lactose utilization (*lac*) operon of *E. coli*, three genes necessary for the successful utilization of the **disaccharide** lactose, a common sugar found in milk, are arranged in a linear fashion on the chromosome (see Figure 1). The *lacZ* gene, which encodes the lactose-degrading enzyme, β -galactosidase, is directly followed by the *lacY* gene, which encodes a membrane protein, called lactose permease, that allows the entry of lactose into the cell. The *lacY* gene is immediately followed by the *lacA* gene, which encodes the thiogalactoside acetyltransferase enzyme that detoxifies lactose-related compounds that might be toxic to the cell. The linear arrangement of these functionally related genes is a hallmark of an operon. The arrangement is significant because the proteins made from these genes will all be easily turned on in concert, so that lactose metabolism proceeds rapidly and efficiently.

Other Typical Characteristics

The clustered genes of the operon typically share a common promoter and a common regulatory region, called an operator. Gene expression requires

synthesis creation

macromolecules large molecules such as proteins, carbohydrates, and nucleic acids

disaccharide two sugar molecules linked together

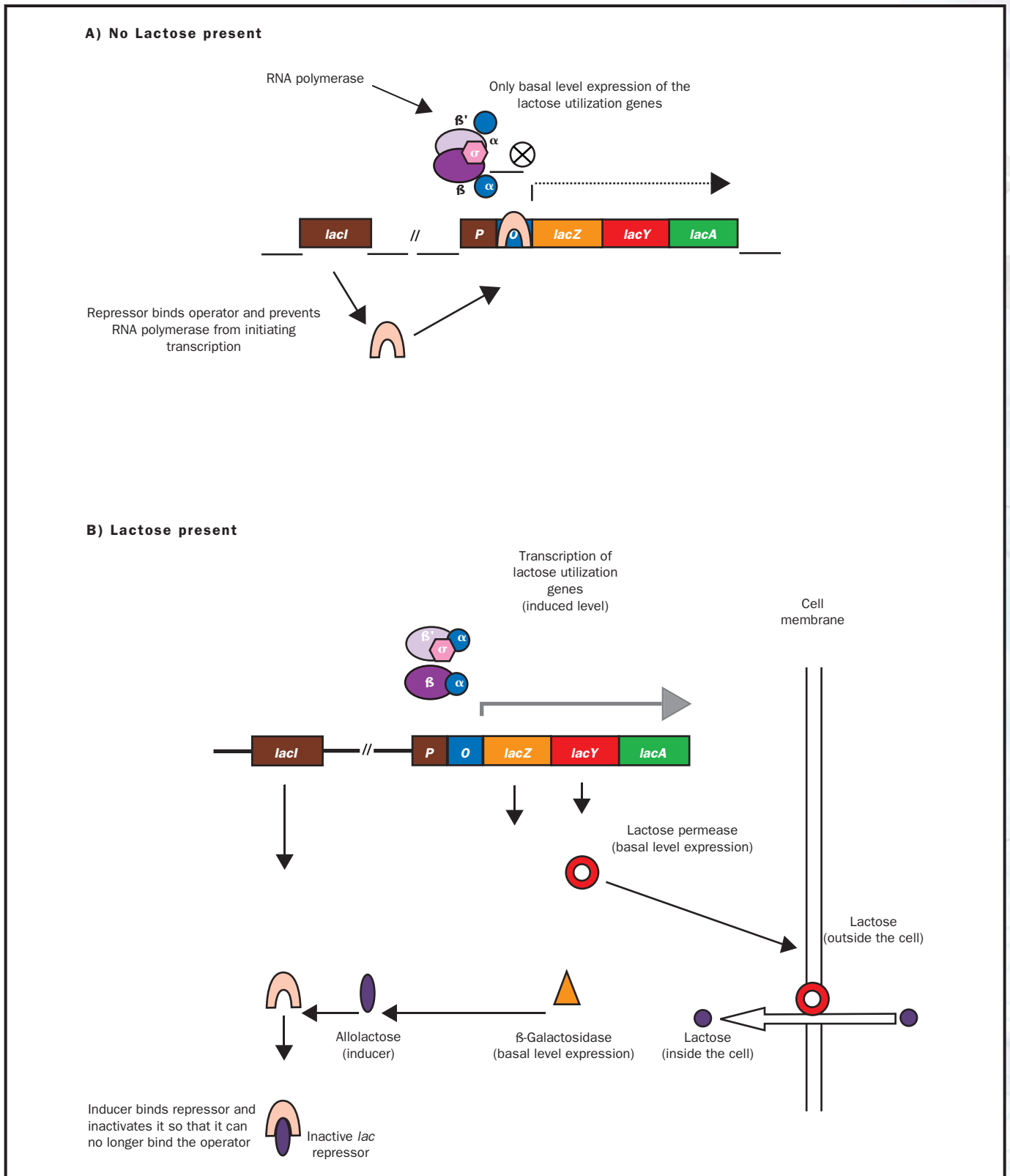


Figure 2. Regulation of gene expression in the *lac* operon. RNA polymerase is represented as a five-subunit protein, with two α subunits, one β subunit, one β' subunit and one σ subunit. The α subunit binds the promoter and dissociates from the polymerase once elongation of the RNA molecule begins. When lactose is present, some is converted to allolactose, the inducer which stimulates transcription of the *lac* operon. Creation of lactose permease increases the cell's ability to take up lactose.

RNA polymerase
enzyme complex that
creates RNA from DNA
template

promoter DNA
sequence to which RNA
polymerase binds to
begin transcription

attenuation weaken or
dilute

allolactose “other lac-
tose”; a modified form
of lactose

the enzyme **RNA polymerase** to transcribe (synthesize an RNA copy of) the gene. This RNA copy is called a messenger RNA (mRNA), which is translated by ribosomes to produce the protein encoded by the gene. In all genes, RNA polymerase begins transcription at a specific site or sequence called the **promoter** (designated “P” in Figure 1). The genes in an operon usually share a common promoter from which the genes of the operon are transcribed.

Operons almost always contain a common promoter region, but not all operons contain only a single promoter. For example the *E. coli* operon for galactose utilization (*gal*) contains two promoters. One of these promoters is active in the presence of glucose, and the other is not (both glucose and galactose are sugars). Some operons, like the *trp* and isoleucine-valine (*ilv*) operons, both from *E. coli*, also have internal promoters that allow the expression of some but not all of the genes in the operon. (Isoleucine and valine are amino acids.)

Operons also have one or more control regions, called operators, that mediate the expression of the genes in the operon (the operator is designated “O” in Figure 1). Like a promoter, an operator is a site on the DNA, but it does not bind with RNA polymerase. Operators function in one of two ways. They can contain DNA sequences that specifically bind particular proteins. Once bound onto DNA, these proteins can prevent the expression of the operon by interfering with the action of RNA polymerase, as in the case of the *lac* repressor. Other proteins bound on other operons can greatly enhance the expression of the operon, as in the case of the AraC protein. Operators can thus prevent or facilitate gene expression.

Instead of acting as target sites for DNA-binding proteins, operators also act as the sites of regulation by **attenuation**. Amino acid biosynthesis operons such as *trp* are usually regulated by attenuation. In such operons the operator provides both a start site for transcription and a ribosome-binding site for the synthesis of a short leader peptide. Through a clever mechanism, the presence of sufficient amino acid in the cell causes the ribosome to disrupt transcription. When the supply of the amino acid is low, transcription of the operon continues without interruption. In this way, if the proteins coded for by the operon genes are needed to synthesize amino acids, then early transcriptional termination does not occur. If they are not needed, because the amino acid is already present, then early termination ensues. This prevents the wasteful production of unnecessary proteins.

The genes of an operon also show a common mode of regulation. The clustering of the genes of an operon and the related functions of these genes requires a mode of regulation that equally affects all the genes of the operon. In the case of the *lac* operon of *E. coli*, the product of the *lacI* gene is a DNA-binding protein that specifically binds to the *lac* operator and prevents RNA polymerase from initiating transcription of the lactose utilization genes from the promoter. Therefore, in the absence of lactose, the lactose utilization genes are only expressed at a very low basal level (see Figure 2A). This low level of expression allows synthesis of a few lactose permease molecules, which permit the entry of lactose into the cell when lactose is present, and a few β -galactosidase molecules, which metabolize lactose or convert it to **allolactose**.

Allolactose is the inducer of the lac operon, acting as a signal that lactose is present. Allolactose binds to the repressor protein, changing its shape in such a way that the repressor can no longer bind to the operator. This allows RNA polymerase to effectively initiate transcription from the *lac* promoter (see Figure 2B).

Transcription of an operon generates an mRNA transcript of all the genes contained within the operon. Ribosomes can translate this single mRNA to generate several distinct proteins. In the case of the lac operon, transcription produces an mRNA molecule that is translated by ribosomes to generate β -galactosidase, lactose permease, and thiogalactoside acetyltransferase. Messenger RNA molecules that encode more than one gene are called polycistronic mRNAs. The common regulation mechanism determines when each polycistronic mRNA is synthesized. This is the main means by which operons commonly regulate the expression of one or more functionally regulated genes. **SEE ALSO** DNA; *ESCHERICHIA COLI* (*E. COLI* BACTERIUM); GENE; GENE EXPRESSION: OVERVIEW OF CONTROL; PROTEINS.

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Overlapping Genes

Overlapping genes are defined as a pair of adjacent genes whose coding regions are partially overlapping. In other words, a single stretch of DNA codes for portions of two separate proteins. Such an arrangement of genetic code is ubiquitous. Many overlapping genes have been identified in the genomes of **prokaryotes**, **eukaryotes**, **mitochondria**, and viruses.

For two genes to overlap, the signal to begin transcription for one must reside inside the second gene, whose transcriptional start site is further “upstream.” In addition, the “stop” signal for the second gene must not be read by the ribosome during **translation**, using the RNA copy of the gene. This is possible because RNA is read in triplets, meaning that it can contain three separate sequences that can be “read” by the cell’s protein-making machinery. Such sequences of nucleotide triplets are called reading frames, and they are different in the RNA transcripts of the overlapping genes.

Overlapping genes enable the production of more proteins from a given region of DNA than is possible if the genes were arranged sequentially. Indeed, for the bacteriophage PhiX174, overlapping of genes is necessary. The amount of DNA present in the circular, single-stranded DNA genome of this virus would not be sufficient to encode the eleven bacteriophage proteins if transcription occurred in a linear fashion, one gene after another.

The genome economy afforded by overlapping genes extends to the human genome. The recently completed sequencing of the human genome has revealed between 30,000 and 70,000 genes. Yet evidence suggests that the human genome encodes 100,000 to 200,000 proteins. At least part of

prokaryotes single-celled organisms without a nucleus

eukaryotes organisms with cells possessing a nucleus

mitochondria energy-producing cell organelle

translation synthesis of protein using mRNA code



the information for the extra proteins may come from the presence of hitherto undiscovered overlapping genes, although more may come from alternative splicing of exons in a single gene.

In algae called *Guillardia*, a structure called a nucleomorph contains only about 500,000 base pairs of DNA, a very small genome, yet produces almost 500 proteins. Part of the efficient packaging of the genome is due to 44 overlapping genes. A nucleomorph is a remnant of a nucleus from an ancient eukaryotic organism that became incorporated into the algae.

One consequence of overlapping genes is to reduce the tolerance for mutation. Virtual experiments conducted within the past several years using a software system called Avida have indicated that overlapping reduces the probability of accumulating so-called neutral mutations in a gene (mutations that have no effect). Neutral mutations are unlikely with overlapping genes, because the mutation must have no effect on two genes with different reading frames.

The evolutionary origin of overlapping genes is not yet clear. Recent research indicates that they may have arisen due to several mutational events. These may include the loss of a signal to stop the transcription process in a gene and a shift in the reading sequence of the genetic components. SEE ALSO ALTERNATIVE SPLICING; DNA REPAIR; GENE; GENETIC CODE; MUTATION; TRANSCRIPTION.

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Patenting Genes

A patent is a legal right granted by the government that gives the patent-holder the exclusive right to manufacture and profit from an invention. While naturally occurring substances in their natural form are not patentable, a very wide range of biological materials have been the subject of patents. In 1980 the U.S. Supreme Court decision in *Diamond v. Chakrabarty* indicated that "anything under the sun made by man" is patentable. Under certain conditions, patent protection is available for genetic information, plants, non-human animals, bacteria, and other organisms.

For naturally occurring substances, patent protection can only be obtained if someone has changed the substance so that it is no longer the same as it is found in nature. For genes, this means that the particular gene of interest must be isolated from other genetic material, such as a chromosome. Although the *Chakrabarty* decision involved bacteria that were mod-

ified using recombinant DNA technology, this case has been viewed as mandating the patentability of other living organisms; nonnaturally occurring, nonhuman multicellular organisms such as transgenic animals, genetic materials, and purified biologically produced compounds such as enzymes. Other cases, such as the 1977 *In re Application of Bergy* and 1979's *In re Kratz*, have provided further support for the patentability of various biological materials, including cells, proteins, and organisms.

Patent Requirements

Although isolating the gene from other genetic material renders the gene “made by man,” various other requirements must be met in order to gain patent protection. For example, in order to be patentable the gene must have a substantial and credible use. Thus, a patent on a gene would not be allowed if the only use described in the patent application was for the use of the gene in some area that was totally unrelated to the function of genetic sequence.

Patents may be obtained for specific DNA sequence information as well as for RNA and amino acid sequences, and for the use of these sequences in various methods. For example, some patents are directed to the use of genetic information in tests to diagnose disease or in test compounds that might be useful to fight disease. Other “methods” patents include the use of the genetic information for tests to identify people with a predisposition to acquiring a certain disease. Other patents are directed toward gene therapy to replace defective genes.

Patent laws apply to gene patents in the same way that they apply to mechanical inventions. In the United States, there is a “first to invent” standard that must be met. This standard means that whoever first invents, discovers, purifies, or isolates a gene is entitled to all patent rights arising from that invention. The law specifies that the invention must have been made by the individual submitting the application. It also requires that the genetic information be “novel.” This means that it cannot have been described in a printed publication more than one year prior to the filing of the patent application. Further, the genetic information cannot have been known or used by someone other than the inventor more than one year prior to the invention by the person who has filed the patent application. If the gene is newly described and has not been publicly disclosed, the novelty requirement is usually easily met. Unlike the United States, however, most countries do not have a one-year grace period between the disclosure of the gene and the filing of a patent application. In those countries it is therefore much more difficult to meet patent law requirements.

To qualify as patentable, genetic information must also be “unobvious.” This means that the gene cannot be an obvious modification of something that is already known. It is usually easy to meet this requirement when DNA, RNA, or amino acid sequences are involved. However, complications sometimes arise when variants of the genetic information are already known.

The patent application must also provide a description of the “best mode” contemplated for making and using the gene. In addition, there are “written description” and “enablement” requirements. These requirements mean that the application must include a description of the gene, its function, and use that is sufficiently thorough to enable someone “skilled in the art” to reproduce the invention. In some cases, meeting the enablement



requirement requires that the inventor provide a deposit of the biological material that is being patented. For example, an applicant seeking to patent genes produced using recombinant DNA technology would have to provide a sample of the genes to a biological materials patent depository, and would have to guarantee that the material will be available for the life of the patent.

The Patent Process

Once the patent application is filed, the application is assigned to a patent examiner, who then has the responsibility of reviewing the application and determining whether the invention is entitled to patent protection. Often, the examiner initially rejects the claims that indicate what the inventor wants to protect. This occurs when the examiner finds these claims to be too broad or unfounded. This leads to a procedure known as the “prosecution” of the application, in which the examiner and the patent attorney or patent agent attempt to resolve the problems. If they are successful, a patent will be issued; otherwise, the examiner will refuse allowance of the patent. If the patent is denied, the inventor can then either appeal the decision or simply abandon the efforts to gain patent protection.

There are various other considerations in filing for patent protection in the United States and other countries. For example, in the United States there is a “duty of candor.” This means that anyone filing a patent application must provide truthful information to the examiner, along with copies of all publications and other materials that might be relevant to the examination of the application. Failure to meet these honesty requirements can result in the invalidation of patents, the disbarment of the patent attorney or patent agent involved, or both.

In addition to patent protection in the United States, many inventors apply for protection in other countries. Most countries, including Japan, Mexico, Canada, Australia, and the nations of Western Europe, recognize genes as being patentable subject matter. However, these other countries have different standards for patentability than those followed in the United States, and many do not allow patents for medical devices. Although there are some patent attorneys who are licensed to practice both in the United States and abroad, most inventors employ the services of foreign patent attorneys who are familiar with the patent laws and rules in the foreign country of interest. SEE ALSO ATTORNEY; BIOTECHNOLOGY ENTREPRENEUR; LEGAL ISSUES; TRANSGENIC ANIMALS.

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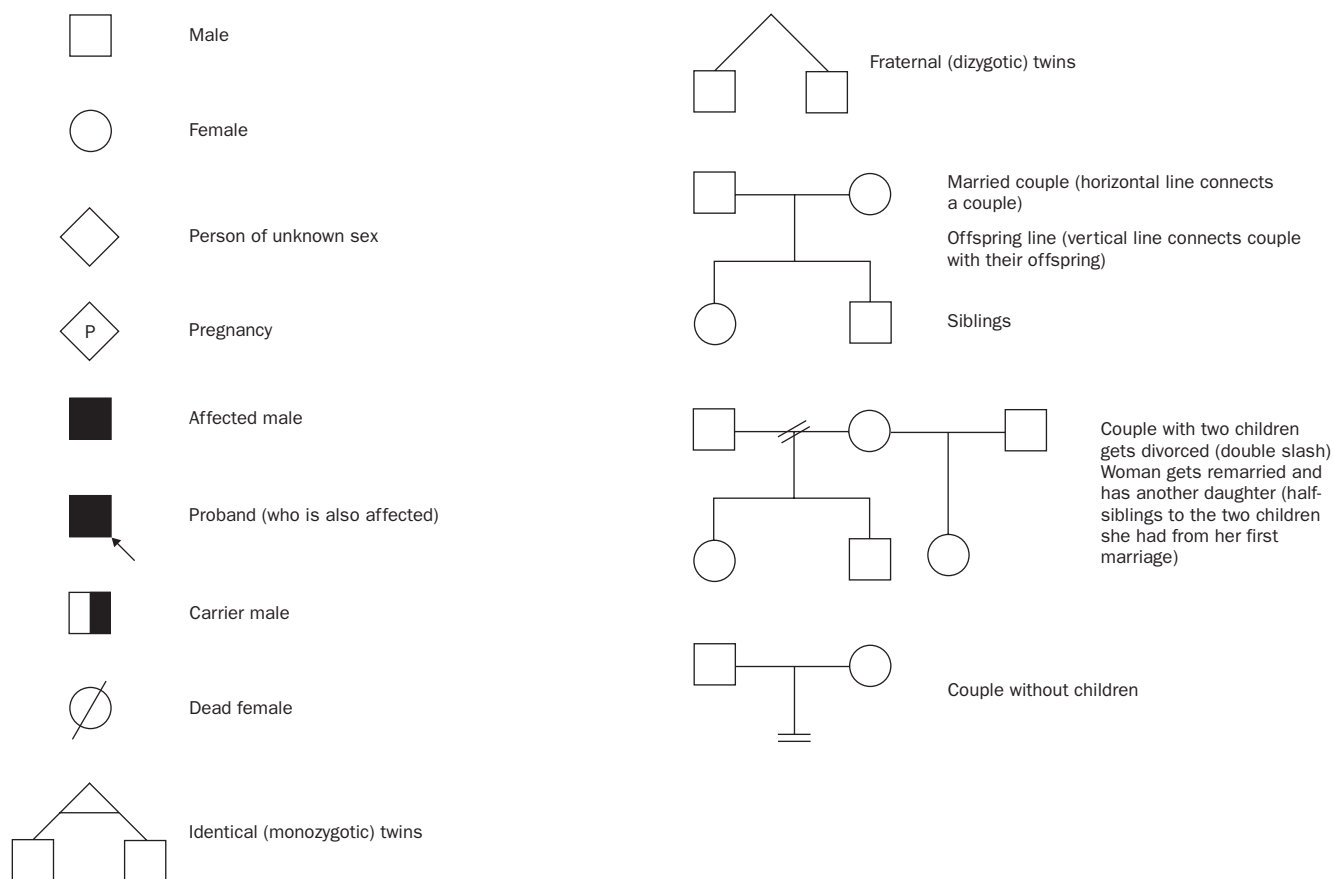
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Pedigree

One of the most important tools used by genetic professionals is the pedigree, a pictorial description of a family tree. A complete pedigree provides

PEDIGREE SYMBOLS



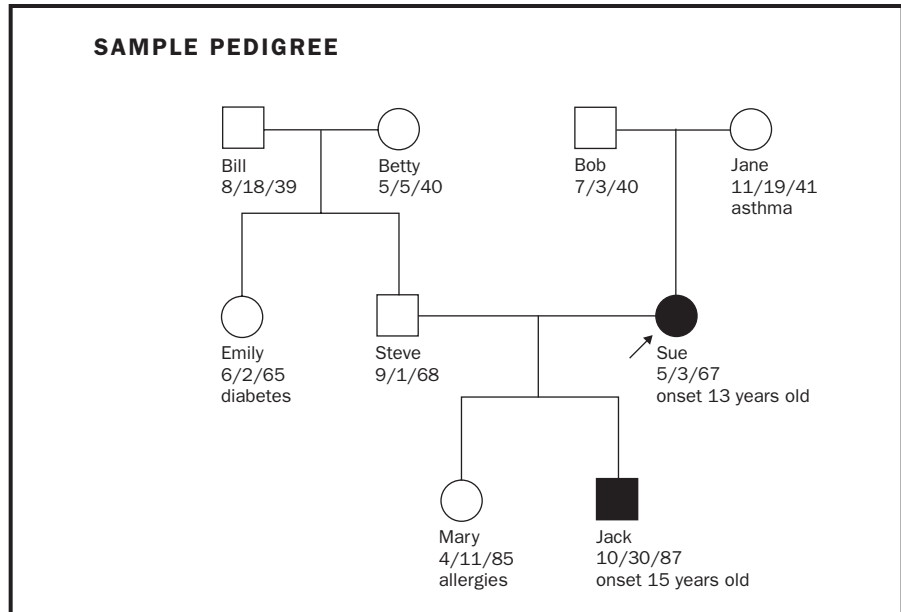
information about the biological relationships of individuals in the family, their medical history, the pattern of inheritance of a genetic disorder in the family, variable expression of the disorder, which family members are at risk, fertility of individuals (including pregnancies, miscarriages, and stillbirths), and family members who are dead. Physicians sometimes refer to a pedigree as a “genogram.” However, genograms usually contain more social information about family relationships than a traditional pedigree used by geneticists and genetic counselors. For example, a genogram can show a teenage child who has a poor relationship with a parent or an individual estranged from the family.

Pedigree symbols used to indicate familial relationships.

Use of Pedigrees

Pedigrees can be used in the clinical setting, such as genetic counseling sessions or genetic evaluations, or in genetic research. By analyzing how many family members have a genetic disorder, how these individuals are related, and the sex of the affected individuals, it is often possible to determine the inheritance pattern of the genetic disorder in the family. Together, the inheritance pattern and an accurate diagnosis help the genetic professional provide accurate risk information to the family. This includes risk information for future pregnancies or relatives who are currently unaffected, but who are at risk for developing the disorder based

In this sample pedigree, Sue is the proband, or person first coming to the attention of the geneticist. Birth dates and relevant medical history is also included.



on family history information. Genetic testing options, if available, can then be offered to those at risk.

The pedigree is also a standard tool used by researchers. For example, in studies aimed at identifying genes that cause human genetic disorders, researchers must collect detailed information on relatives participating in the study, particularly those relatives who are affected with the disorder. Researchers compare the genes of affected individuals with the genes of those who did not inherit the disorder to identify the specific genes responsible. In other studies the disease-causing gene is known, and researchers study the gene **mutation(s)**. A pedigree can help identify which family members should be included in mutation analysis, as only those family members who are affected or are at-risk could carry a mutation. Researchers can also pictorially show laboratory data, such as **genotypes** or **haplotypes**, on the pedigree.

mutation change in DNA sequence

genotypes sets of genes present

haplotypes sets of alleles or markers on a short chromosome segment

Terminology

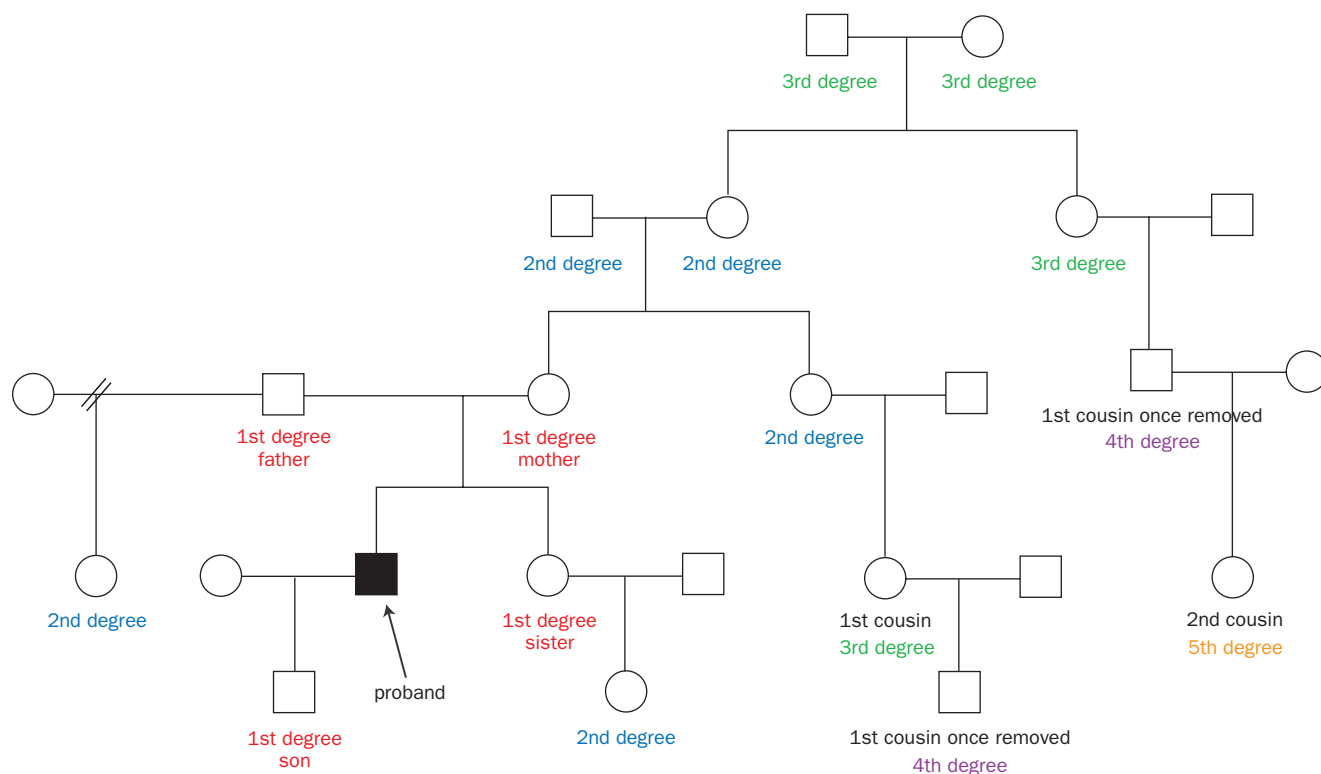
Standard symbols are used to draw pedigrees. For example, males are represented by squares and females by circles. The individual who brought the family to the attention of the medical professional or researcher is called a “proband” and is identified on the pedigree by an arrow pointing towards the symbol for that individual. This individual usually has the disorder of interest to the researcher or physician.

The informant is the individual who was interviewed to obtain the pedigree. The informant may or may not be the same person as the proband. The name of the informant, date the pedigree is drawn, and interviewer are also noted on the pedigree. So, too, is the founder of the pedigree, who is the first family member known to have had the disorder.

Drawing and Recording Pedigrees

Pedigrees are hand-drawn or created using special computer software. The standard pedigree typically includes at least three generations, with each gen-

DEGREES OF RELATIONSHIP: PROBAND AS REFERENCE



eration arranged horizontally and connected to the other generations by lines. Family members who have the genetic disorder in question are colored-in or shaded. Unique symbols represent carriers, miscarriages, people of unknown sex, twins, and other categories of individuals. Furthermore, different patterns within the pedigree symbol may represent variable expression (variation in symptoms of individuals with the same disorder). For example, in a pedigree showing myotonic dystrophy, a shaded upper-left quadrant of the male or female symbol may represent cataracts, but a shaded right-lower quadrant may represent heart problems.

The process of collecting the family history used to draw the pedigree includes interviewing the informant and asking questions in three general categories. The detailed information about family members is recorded under the symbol for that individual. First, the interviewer asks standard questions about the family members, resulting in a skeleton picture of the family. These questions collect identifying information about each family member, such as their names, names of their parents, siblings, and children, dates of birth, dates of death, cause of death, and pregnancies. Detailed information is also included about general health problems, such as cancer and heart disease, and their specific symptoms, onset, and age at diagnosis. The interviewer may also ask about the ethnic background of family members and any possible consanguineous relationships (relationships between individuals who are related to each other by blood, such as first cousins).

Second, the interviewer may ask general questions to identify common genetic syndromes in the family, such as birth defects, mental retardation,

Degrees of relationship shown within a simplified pedigree. The proband is the individual first coming to the attention of the geneticist.

vision and hearing loss, and common genetic disorders found more prevalently in certain ethnic backgrounds. For example, cystic fibrosis is relatively more common in people of Western European decent, whereas sickle cell disease is more common in those of African decent. If any common genetic syndromes are identified, they are then discussed during the genetic counseling appointment.

Lastly, the interviewer may collect targeted information about the specific genetic disease for which the family was referred. Information is gathered on the symptoms of the disease, age of onset, and age at diagnosis. These targeted questions help genetic counselors and researchers identify the typical course of disease in the family, disease severity, possible variable expression, reduced penetrance, or genetic anticipation. Because accurate diagnoses are essential for accurate genetic counseling and research studies, all diagnoses should be confirmed by medical records. Further, pedigree updates can be obtained over time during follow-up genetic counseling sessions or research interviews.

Confidentiality

Pedigrees contain very personal and identifying information about families. It is therefore essential that pedigrees be kept confidential. This is important not only in the genetic counseling clinic or research group, but also within the family. It is unethical to share information given about family members with other family members, insurance companies, or other professionals not involved in the genetic counseling sessions or research study. Information about individuals should only be released with proper written consent by that individual. SEE ALSO DISEASE, GENETICS OF; GENETIC COUNSELING; HUMAN DISEASE GENES, IDENTIFICATION OF; INHERITANCE PATTERNS.

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Pharmaceutical Scientist

The role of the pharmaceutical scientist in drug discovery and development is highly varied. Duties range from the synthesis of novel compounds designed to alter disease processes, to the formulation of these compounds into a tablet or capsule, to the development of assays (tests) to measure the drug and its **metabolites** in the body, to the testing of compounds for their effects in animals and humans. In addition, scientists in this industry pursue more basic questions, such as which genes or processes are critical in disease, as well as better ways to diagnose disease and predict clinical outcomes of treatment strategies. While the pharmaceutical industry employs thousands of chemists and biologists, the skills needed to work in the phar-

metabolites molecules involved in a metabolic pathway

maceutical industry are much broader and are being substantially changed by the infusion of genetics and genomics into the drug development process. Therefore those interested in a scientific career in the pharmaceutical industry should seriously consider training in the important areas of genetics and **genomics**.

The drug development processes can be divided into three major sections: research, where compounds are synthesized and tested against potential drug targets and for activity in animal models of disease; preclinical safety, where the compounds are analyzed for their potential toxicity in the laboratory and in animal model studies; and manufacturing of clinical-grade material and testing in human clinical trials. Individuals with a range of skills in chemistry, biology, manufacturing, and clinical sciences are very important. However, it is not limited to these areas, since the pharmaceutical industry employs most types of scientists. The increasing amount of genome sequence available at the present time has generated a need for individuals trained in bioinformatics. These scientists use computational methods to answer biological questions, particularly methods involving massive amounts of data produced by the field of genomics. In addition, scientific expertise is needed in many of the support areas of the drug development process, including the business, legal, and regulatory aspects. As a consequence, the training, skills, and qualifications needed for work in the industry are very broad and varied.

Beginning at the technical level with a college degree, there is a variety of entry-level positions in all areas of drug development. Opportunities also exist for individuals to work on postgraduate degrees within a pharmaceutical company, and there are many options for conducting postdoctoral research throughout the industry. Naturally, the work environment varies as much as the types of positions. Pharmaceutical scientists are typically based in the laboratory, manufacturing facility, or office. Those working on clinical trial design will typically work from corporate offices and then implement the patient treatment with investigators at universities or clinics, rather than actually conducting the patient research themselves.

Due to the wide variety and types of positions in this global industry, salary ranges are very broad. Historically, pharmaceutical scientists receive competitive salaries and they may also receive cash or stock bonuses. The pharmaceutical industry is oriented toward extremely high-quality research that leads to new treatments for people in need. Successful scientific work leads to useful drugs that benefit patients as well as the company that develops them. Therefore, the success of pharmaceutical scientists is tied to the degree to which their work benefits patients and to the degree of financial success achieved by the company as a whole.

A career as a pharmaceutical scientist can be exceptionally rewarding. It provides the professional with an opportunity to participate in a team that seeks to discover useful new drugs. There is satisfaction in knowing that, when approved and sold on the market, such a discovery can help millions of people for years to come. In addition, pharmaceutical scientists have the opportunity to work in cutting edge areas, using new methods for studying genetics in clinical trials. One such field of study, called pharmacogenetics, examines the reasons that individuals have different responses to the same drug. This area of study is expected to greatly improve the understanding of how drugs work and enable physicians to prescribe them to those who

genomics the study of gene sequences



are most likely to benefit, while minimizing the risk of adverse reactions. SEE ALSO BIOINFORMATICS; PHARMACOGENETICS AND PHARMACOGENOMICS; PHYSICIAN SCIENTIST.

Kenneth W. Culver and Mark A. Labow

Pharmacogenetics and Pharmacogenomics

genome the total genetic material in a cell or organism

The complete sequencing of the human **genome** in 2000, along with new technologies, such as DNA microarrays, for analyzing human genes on a genome-wide scale, provides scientists with the tools to study the molecular basis of diseases on a level and scale that previously had not been possible. Pharmacogenomics is a biomedical science that aims to use this knowledge to tailor drug therapies based on patients' individual genetic makeup. Doctors hope to use pharmacogenomics to develop safer and more effective medical treatments. For some diseases, this promise has already been realized.

Molecular Interactions and Drug Effectiveness

heritable genetic

Pharmacogenomics is a branch of pharmacogenetics, a science that deals with the **heritable** traits responsible for the individual differences in the ways people respond to drugs. It is remarkable, considering the myriad of medications we have today, how little we understand about how most of them actually work. There are many factors that influence the effectiveness of a particular drug, including how the drug enters the body's cells, how rapidly it is degraded by metabolic **enzymes**, and how it interacts with target molecules in the body, such as drug receptors.

enzymes proteins that control a reaction in a cell

Consider, for example, a common general anesthetic drug, called succinylcholine. In the 1950s doctors noticed that some patients suffered prolonged respiratory apnea (difficulty breathing) after being treated with succinylcholine. This syndrome was found to be caused by mutations in a gene for the enzyme butyrylcholinesterase. Normally butyrylcholinesterase in the blood degrades succinylcholine, and the anesthetic effects of the drug wear off with time. But in patients with mutations that inactivate or weaken butyrylcholinesterase, the anesthetic persists in the body, causing the dangerous side effect.

alleles particular forms of genes

Cytochrome P450 is a member of a large family of enzymes that inactivate more than half of all drugs. There are many different **alleles** of cytochrome P450. Some alleles are very inefficient at inactivating drugs. In individuals with these alleles, termed "poor metabolizers," drugs can accumulate in the body to levels that produce toxic effects. In contrast, some people have extra copies of cytochrome P450 genes and produce excessive levels of the enzyme. In these individuals ("ultrarapid metabolizers"), drugs become inactivated so rapidly that they may not accumulate to the concentrations needed to be effective.

genotype set of genes present

Adverse drug reactions such as the examples just discussed account for over one hundred thousand deaths each year in the United States. If a physician can determine the **genotype** of patients with respect to the genes that



Stephen Liggett uses this cell harvester to perform research in his laboratory at the University of Cincinnati. Dr. Liggett works towards understanding better how genes and drugs interact.

encode these proteins, then he or she may be able to prescribe the most appropriate dosage for a particular drug, or a better-suited drug.

A major obstacle, however, is that so many different proteins affect each drug. Some are well characterized, but most are completely unknown. This is where pharmacogenomics promises to revolutionize medicine. With the sequence of the entire human genome having been determined, and with the development of modern gene analysis technologies, scientists may now be able to pinpoint every gene that influences the effectiveness of any drug. Physicians would then be able to genotype their patients. For example, they could determine whether a patient has P450 alleles that make him a poor metabolizer or an ultrarapid metabolizer.



Not only can pharmacogenomics provide information about the best drug therapy for patients, but it can also be used to predict whether a person is predisposed to contracting a heritable disease. Mutant alleles of many genes have been shown to predispose people to diseases such as breast cancer, Alzheimer's disease, and Huntington's disease. If doctors can identify such mutant alleles in patients long before any sign of disease becomes apparent, they may be able to treat the disease better when it first appears or even prevent it before it strikes.

Of course, this powerful technology carries with it many ethical questions: If you carried a gene that gave you a moderate probability of eventually contracting a fatal disease, should you be told? What if there were no treatments for the disease? Who should have access to a patient's genetic information? If a health insurance company finds out that a person has a set of genes that predispose her to a disease that is costly to treat, should it be allowed to deny her insurance coverage?

Genetic Diagnoses That Can Improve Treatments

In addition to the complex interactions that drug compounds can have with molecules in the body, there are other reasons why some patients experience different responses to drugs. Two people that appear to have the same disease may actually have different diseases that may not respond to the same drug treatment. Modern genomic analysis methods, such as microarray gene expression profiling, can distinguish two diseases that, by all other clinical and diagnostic methods, appear to be identical.

An example is diffuse large B-cell lymphoma (DLBCL), the most common form of non-Hodgkin's lymphoma, a cancer of the white blood cells. About 40 percent of patients can be cured by the standard chemotherapy for DLBCL, but 60 percent respond poorly and eventually die. Using microarray technology to gather and compare the gene expression patterns of cancer cells from many different DLBCL patients, molecular biologists have discovered that DLBCL actually comprises two distinct disease forms. Patients with one form are treatable with the standard chemotherapy, while those with the other form are not. Using this kind of genomic information, doctors can now diagnose patients more accurately and put them on the most appropriate drug-therapy regime. Similar genomic diagnoses based on gene expression profiles are being developed for important diseases such as breast cancer and Alzheimer's disease.

Using SNPs to Identify Disease Genes

Of the approximately 3 billion **nucleotides** in each of the two sets of chromosomes in human cells, the vast majority are identical from person to person. On average, though, the genomes of individuals differ from one another by about 1 million nucleotides. This is largely what accounts for the enormous diversity of humans. A nucleotide in a gene from one person that is different from the nucleotide found at the same position of the gene in most other people is called a single nucleotide **polymorphism**, or SNP (pronounced "snip").

Almost 1.5 million human SNPs have been identified and catalogued as part of the human genome sequencing project. Many of these polymorphisms are within the protein-coding or control regions of genes and may

nucleotides the building blocks of RNA or DNA

polymorphisms DNA sequence variant

contribute to particular diseases, to a predisposition to a disease, or to adverse drug reactions. By comparing the SNP patterns of many different people, geneticists can infer whether a particular SNP (and therefore the gene it is in) is correlated with a disease or adverse drug reaction. Once a correlation is found, doctors can determine if their patients have the SNP in their genomes, to test for the likelihood of contracting the disease or experiencing the adverse reaction. SEE ALSO DNA MICROARRAYS; GENOMIC MEDICINE; POLYMORPHISMS.

Paul J. Mublrud

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Physician Scientist

According to Webster's II New College Dictionary, a physician is "a natural philosopher, a person skilled in physic, or the art of healing; one duly authorized to prescribe remedies for, and treat, diseases; a doctor of medicine." A scientist is one who is "learned in the observation, identification, description, experimental investigation, and theoretical explanation of phenomena." The combination of these definitions precisely describes a physician scientist.

A physician scientist typically has two very different aspects to his or her career. All of the traditional duties of a physician are performed, including caring for patients. The usual responsibilities of conducting rounds in the hospital and treating patients in clinic are also carried out. The physician may be a generalist who sees all types of illnesses, or a specialist whose practice focuses on a certain organ system. An example would be a nephrologist or renal doctor, who cares for people with kidney diseases. In addition to these conventional medical duties, however, the physician scientist typically maintains a laboratory, designs experiments, and supervises a staff of technicians who carry them out.

There are commonalities between both careers. They both characteristically take a great deal of self-motivation, and both require a commitment to continuing education in order to stay abreast of advances in science and medicine. And both the physician and the scientist are responsible for other individuals. Patients rely on the physician for honesty and competent care; technicians rely on the scientist for supervision, support, and employment. In either capacity, the physician scientist may also be required to teach students. The physician uses the knowledge of his or her particular field to enhance his science career and vice versa.

To pursue a career as physician scientist, a candidate completes college, medical school, and a residency program, as is required to earn a medical degree. He or she may then subspecialize in a field by completing a fellowship program. From college to the conclusion of such a fellowship takes



at least thirteen to fifteen years. The physician scientist may or may not have a Doctorate of Philosophy (Ph.D.) specifically as a scientist. Such training characteristically includes approximately five years of graduate school and usually some postdoctoral training as well. Occasionally, physician scientists may replace such formal education with “on-the-job” training in the laboratory, but in such cases they supplement the scientific skills and knowledge they developed in medical school with courses and mentorship from a senior scientist.

The physician scientist may work in a number of different environments, including hospitals, universities, the government, or industry. Because physician scientists do not usually see as many patients in a year as a typical doctor, they are usually salaried, and their salaries are usually covered by grants. Thus, for physician scientists who are employed in a hospital affiliated with an academic institution, part of their income may be based on the number of patients in their care, or how many months of the year they perform rounds in the hospital. To secure grants, physician scientists submit applications to a number of organizations, including the government, charities, and private companies. Their applications provide a description of the proposed research, the questions they want to ask and answer with their research, and how they plan to find these answers (their experimental plan). Alternatively, a physician scientist may work for a private research company such as a pharmaceutical firm. The salary range for physician scientists is from approximately \$70,000 to more than \$200,000, depending on experience and productivity. Productivity is usually measured by how much grant funding has been obtained, how many scientific papers have been published, and recognition by peers. In the industrial sector, productivity will also be measured by how much the research has contributed to successfully bringing products to market.

The combination of physician and scientist results in a very exciting career choice. The physician scientist has the pleasure of taking care of patients and potentially having a direct impact on their future health by the discovery of an important clue to their illness. This is sometimes referred to as conducting science “from-bedside-to-bench-to-bedside.” It also provides variety, because the physician scientist is not only able to practice medicine in a very concrete clinical way, but can also think through a variety of more abstract issues, extending his or her intellect in very different directions. SEE ALSO GENETICIST; PHARMACEUTICAL SCIENTIST.

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Plant Genetic Engineer

“Plant genetic engineer” is a popular term that describes scientists working in any of several fields who manipulate DNA or **organelles** such as **chloroplasts** and **mitochondria** in plant cells. The specific titles of such a scientist can include plant physiologist, plant pathologist, weed scientist, cell biologist, botanist, molecular biologist, plant geneticist, and biochemist. The typical career path is to earn a doctorate in any of these fields, and then to go to work in industry, for the government, or in academia.

Jobs that involve genetic manipulation of plants are also available at the technician level. This requires a bachelor’s or master’s degree in any of the fields listed above. As of 2001, the median salary for an academic life scientist is \$42,000, and for a scientist in industry, \$70,000. Technician salaries begin at about \$25,000. Government salaries are similar to academic compensation.

Scientists work with plant genes in basic research as well as in developing new crops. In basic research, a plant may serve as a model system, enabling researchers to study a fundamental structure or function at the cellular or molecular level. The mustard plant *Arabidopsis thaliana* is the most popular plant model system. Most work on this plant is not to alter or develop a product from it, but to study its basic biology and then extrapolate what is learned to other plant species. Similarly, *Zea mays* (corn) was used extensively in the mid-twentieth century to reveal how chromosomes interact as a cell divides. Today, entire genomes of plants and other types of organisms are being sequenced, which is enabling researchers to work with several genes at a time. The genomes of *Arabidopsis* and several major crop plants are being sequenced.

Some plant scientists manipulate genes to create new variants. Adding a gene from another species forms a transgenic organism. For example, “bt corn” plants harbor a gene from a bacterium that enables plant cells to produce a toxin that kills certain insect larvae that feed on the leaves. “Golden rice” produces beta carotene, a precursor to vitamin A, which is not naturally abundant in the grain portion of this plant. The ability to manufacture beta carotene comes from three genes, taken from daffodils and a bacterium, that specify enzymes that interact in a biochemical pathway, along with genetic instructions to express the genes in the grain, so that it can serve as a vitamin-enriched food.

A plant genetic engineer must be familiar with the characteristics that distinguish plants from other types of organisms. Unlike animal cells, plant cells have tough cell walls, which must be penetrated to reach the DNA. Also, some genetic material resides in the organelles called **plastids**, the largest of which is the chloroplast. The engineer must also be able to regenerate an altered plant cell into a plant, test that plant in a greenhouse, and, finally, see how well it flourishes in a field environment. For example, tomato plants can be given a gene from *A. thaliana* that enables them to grow in very salty water. Developers of such a crop must analyze how the plant that can now grow in **brackish** or salty water will affect other types of plants that normally grow in that environment. Thus, in addition to understanding genetics, molecular biology, and biochemistry, a plant genetic engineer working on an agricultural variant must also have expertise in plant development and

organelles membrane-bound cell compartments

chloroplasts the photosynthetic organelles of plants and algae

mitochondria energy-producing cell organelle

plastids plant cell organelles, including the chloroplast

brackish a mix of salt water and fresh water

